EXPLORATORY DATA ANALYSIS PROJECT REPORT

Homicide Reports, 1980-2014

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1. Introduction

1.1 Dataset

The dataset is the Homicide Reports dataset from Kaggle (murderaccountability/homicide-reports), covering homicide records in the United States from 1980 to 2014. The supplied CSV contains 638,454 rows and 24 original columns. The columns describe the record and reporting agency, location, year and month, crime-solving status, victim demographics, perpetrator demographics, relationship, weapon, victim/perpetrator counts, and record source. In the original CSV, 6 columns are stored as integers and 18 as text/object columns. For this analysis, Crime Solved is the main outcome used to compare solved and unsolved records; it is not a predictive-model target.

1.2 Problem Definition

This project explores how homicide records vary over time, across states, and across selected characteristics of victims and perpetrators. The analysis asks: How did annual record counts change from 1980 to 2014? Which states recorded more cases, and how did their case-solving rates differ? Which weapons were most frequently recorded, and how did solved rates vary by weapon? What patterns appear in perpetrator ages and age differences between victims and perpetrators? These are descriptive questions. The dataset alone does not establish the causes of homicide trends or differences in case-solving rates.

2. Data Exploration

2.1 Dataset Overview

The original file contains 638,454 records and 24 columns, covering 35 years (1980–2014) and 51 state-level labels, including the District of Columbia. Crime Solved contains 448,172 Yes records (70.20%) and 190,282 No records (29.80%). Handgun is the most frequently recorded weapon (317,484 records). The annual count is highest in 1993 at 24,335 records; the count in 2014 is 14,331. The original CSV has no pandas null values, but some fields use codes such as 0 or 998 and category labels such as Unknown to represent unavailable information.

Key data-quality counts in the raw file: perpetrator age is recorded as 0 in 216,327 rows; perpetrator sex is Unknown in 190,365 rows; relationship is Unknown in 273,013 rows; and weapon is Unknown in 33,192 rows. These coded values need to be distinguished from true missing values.

2.2 Initial Observations

Initial exploration showed that annual homicide-record counts peaked in 1993 and were lower by 2014. The solved/unsolved split is approximately 70.2% versus 29.8%. Handguns dominate the weapon counts. A major data-quality issue is that many perpetrator ages are coded as 0 rather than being stored as blank or Unknown; this occurs in 99.66% of unsolved records and 5.96% of

solved records. The fields Perpetrator Sex, Relationship, and Weapon also contain many Unknown labels. These values should not be treated as ordinary age or category values in analyses.

3. Data Cleaning

3.1 Missing Values

The raw CSV contains no pandas-recognized null values, so checking isna() alone does not identify all unavailable information. The age columns were converted to numeric values with errors='coerce'. In the notebook, victim ages coded 998 or higher were changed to missing (974 records), and perpetrator ages below 8 were changed to missing (216,551 coded values; one additional blank value also becomes missing during numeric conversion). The analysis of perpetrator age therefore uses only records with valid ages and excludes the large group coded as 0. Category values such as Unknown were kept in the main dataset and excluded only where a particular visualization required known values.

3.2 Duplicates and Inconsistent Values

The raw CSV has 0 exact duplicate rows and 0 duplicated Record ID values. Column names were standardized to lowercase with underscores to make them easier to use in Python. The age columns were converted from text/numeric representations to numeric values, then to nullable integer types so that missing ages could be represented. The analysis also creates a location field by combining city and state, and a month_num field to order months chronologically. Unknown labels and special numeric codes were treated as data-quality indicators rather than as meaningful ages.

3.3 Final Dataset

The cleaning steps in the notebook retain all 638,454 rows; they recode age values rather than deleting whole records. The 24 original columns are renamed to standardized lowercase names, age fields are converted to numeric nullable integers, and two analysis columns (location and month_num) are added, giving 26 columns in the working DataFrame. Perpetrator ages below 8 and victim ages coded 998 or higher are represented as missing after cleaning. Unknown categories remain in the dataset and are filtered out only for visualizations where they would make the comparison unclear.

4. EDA (with captions)

4.1 visualization

```
...         state  year  Crime_Count
    0     Alabama  1980          513
    1     Alabama  1981          453
    2     Alabama  1982          424
    3     Alabama  1983          372
    4     Alabama  1984          382
    ..        ...   ...          ...
    95    Arizona  2006          509
    96    Arizona  2007          516
    97    Arizona  2008          403
    98    Arizona  2009          370
    99    Arizona  2010          400

    [100 rows x 3 columns]
                   state  year  Crime_Count
    1659  West Virginia  1984           88
    1660  West Virginia  1985           75
    1661  West Virginia  1986          120
    1662  West Virginia  1987          104
    1663  West Virginia  1988          101
    ...             ...   ...          ...
    1754        Wyoming  2010            8
    1755        Wyoming  2011           20
    1756        Wyoming  2012           17
    1757        Wyoming  2013           17
    1758        Wyoming  2014           14
```

Fig :1

Insight :

Grouping homicide records by state and year reveals how crime patterns vary across U.S. states over time. This allows us to identify states with higher homicide counts, compare yearly trends, and detect periods of increase or decline. These comparisons provide a clearer picture of the geographical and temporal distribution of homicide across the United States from 1980 to 2014.

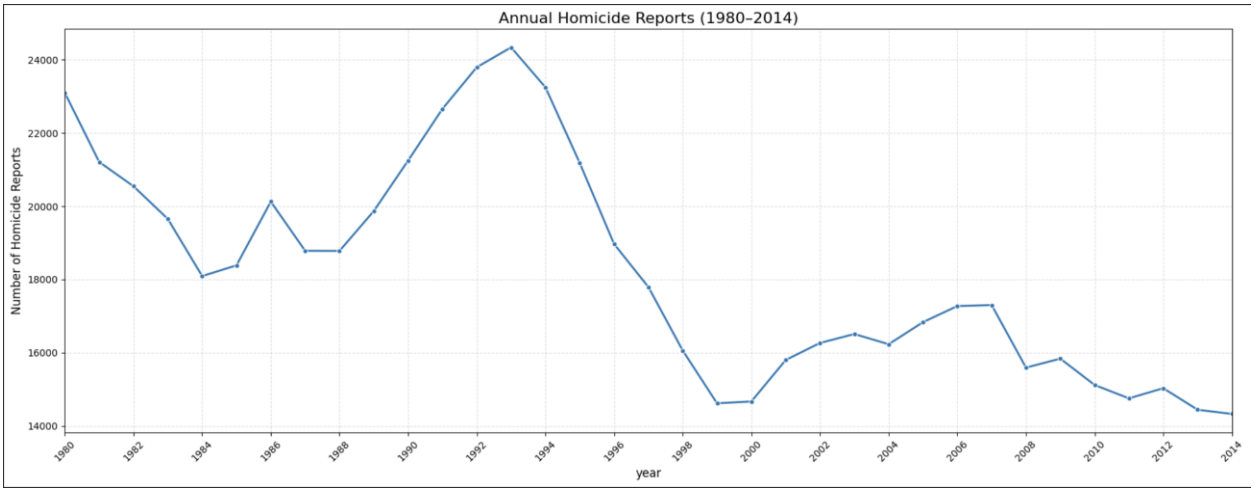


Fig: 2

Insight :

The line chart shows fluctuations in annual homicide-record counts, reaching a peak of 24,335 records in 1993. Counts declined substantially after that peak, reaching 14,331 records in 2014.

The chart describes the trend over time but does not establish what caused the increase or decline.

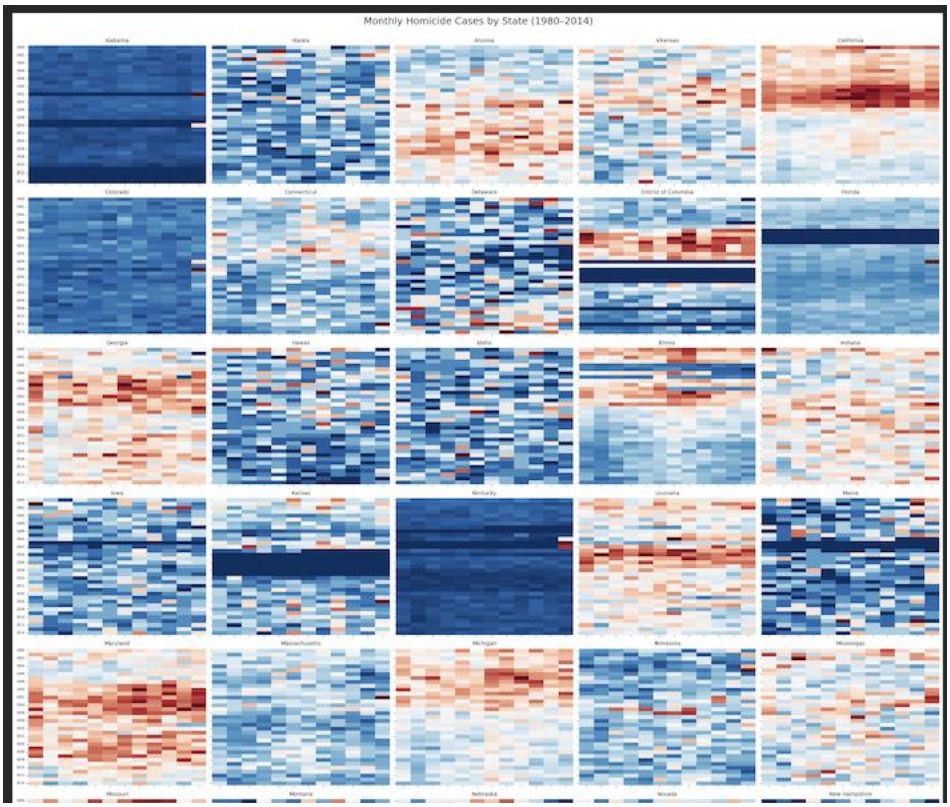


Fig:3

Insight:

The heatmaps compare monthly homicide-record counts across years and states. Nationally, counts were highest around the early 1990s (the busiest month is December 1992, with 2,636 records) and lower in later years. In the state panels, each state is scaled from 0 to 10 using its own busiest month, so the colors show timing within a state and cannot be compared between states. New York, Texas and California peaked around 1991–1993 and Michigan earlier, in 1987; in all four, the yearly average in 1995–2014 was 30–55% below the 1980–1994 average. The colors show recorded counts, not population-adjusted homicide rates. Research links the broader 1990s decline to factors such as the end of the crack cocaine epidemic, increased policing and higher incarceration, but these charts alone cannot confirm any of them as the cause.

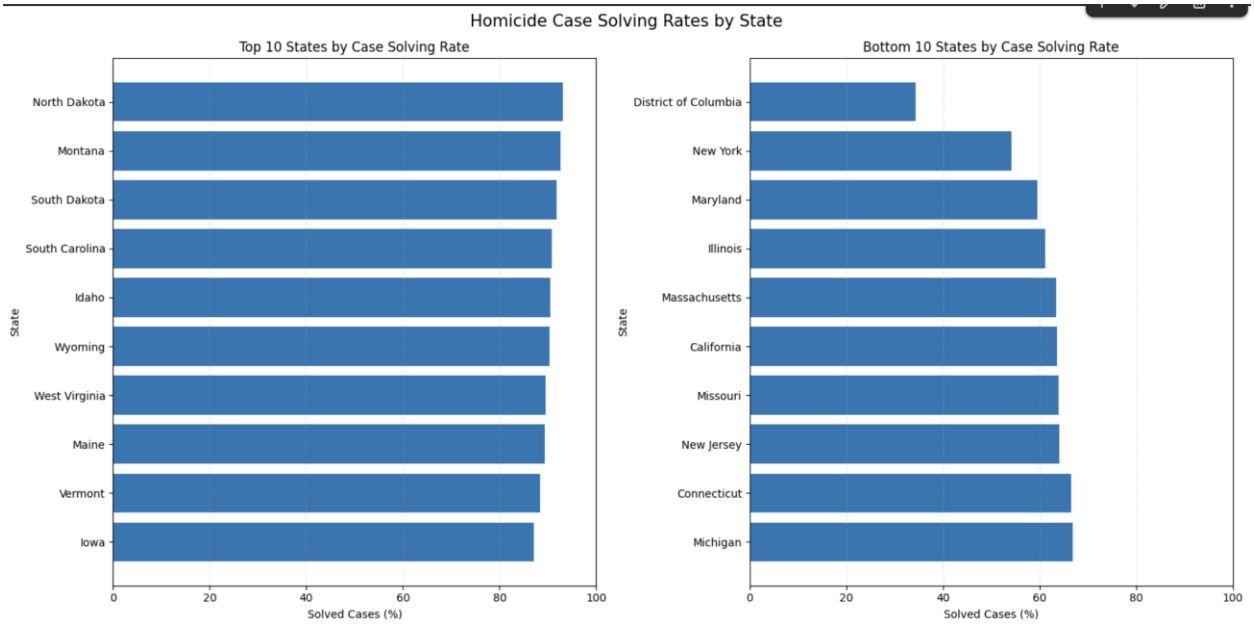


Fig : 4

Insight :

The horizontal bar charts reveal substantial differences in homicide clearance rates across states. North Dakota has the highest rate at 93.18%, while the District of Columbia has the lowest at 34.31%. Notably, California and New York recorded high case counts but comparatively low clearance rates of 63.55% and 54.10%, respectively. This suggests that higher homicide volumes do not necessarily correspond to higher clearance rates.

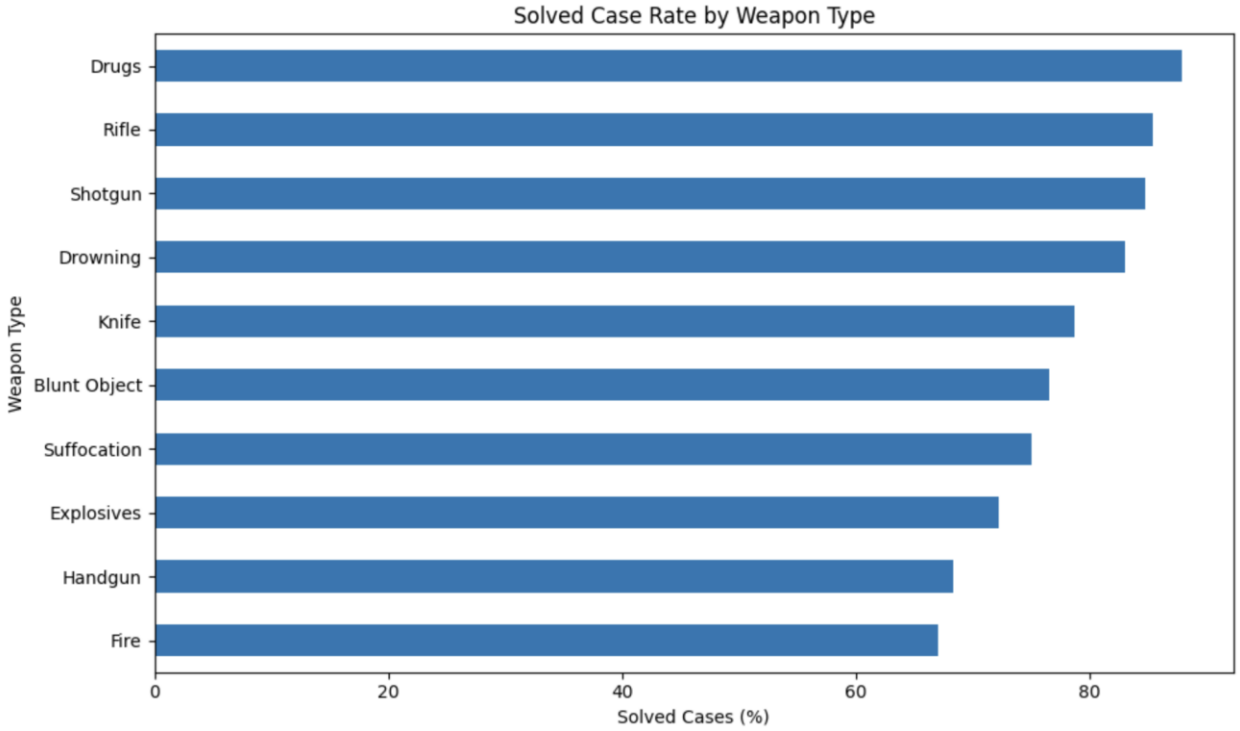


Fig :5

Insight :

Handguns were the most frequently recorded weapon, with 317,484 records and a solving rate of 68.33%. Among weapon types with at least 500 records, Drugs had the highest solving rate (87.91%), followed by Rifle (85.43%) and Shotgun (84.76%). Firearm (51.94%) and Strangulation (54.36%) had the lowest solving rates among these weapon types, although they fall outside the top-10 chart shown. These differences are descriptive and do not explain why resolution rates vary.

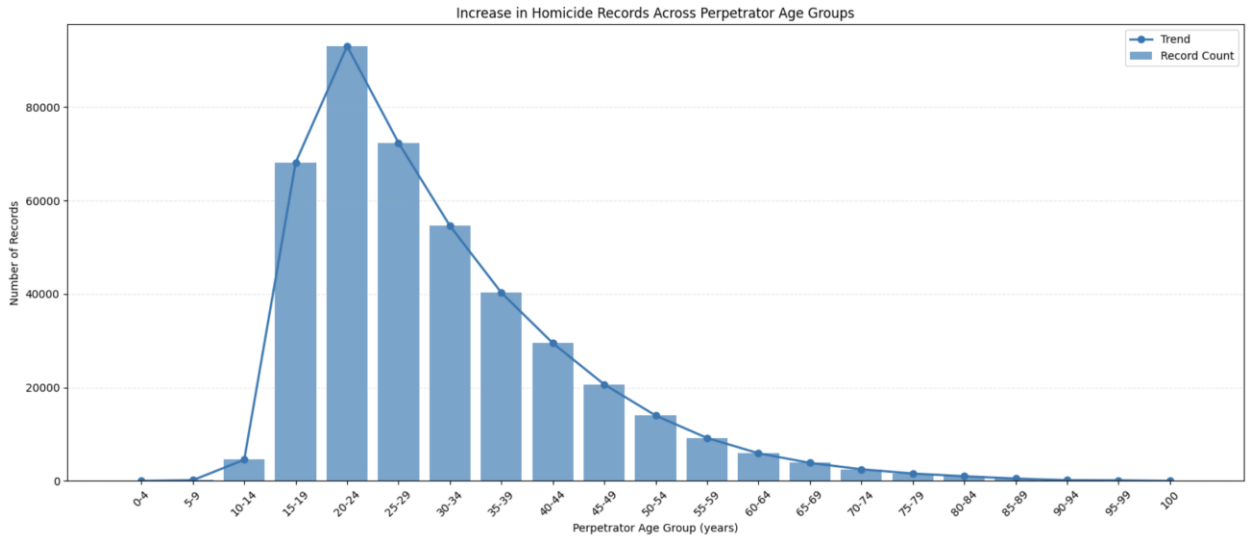


Fig :6

Insight :

The chart shows that homicide perpetrators were most frequently aged 20–24, representing the peak among all age groups. After this peak, the number of cases generally declines as age increases, reaching much lower levels among older age groups. This suggests that young adults accounted for the largest share of recorded perpetrators with known ages. Among the 421,902 records with a known perpetrator age, the 20–24 group has 93,024 records (22%) and the median age is 27. Perpetrators of unsolved cases have no recorded age, so the chart describes identified perpetrators only.

4.2 Relationships Between Variables

The state comparison shows that the number of recorded cases and the case-solving rate do not always move together: states with large case counts can still have lower solving rates. The weapon analysis also shows differences in solving rates across categories. For example, among weapon types with at least 500 records, the recorded solving rate is 87.91% for Drugs and 85.43% for Rifle, while it is 51.94% for Firearm and 54.36% for Strangulation. These are descriptive associations only; they do not explain the reasons for the differences. Age-gap analysis includes only records with valid ages for both the victim and perpetrator.

5. Dashboard Overview (screenshots)

5.1 Dashboard Screenshot

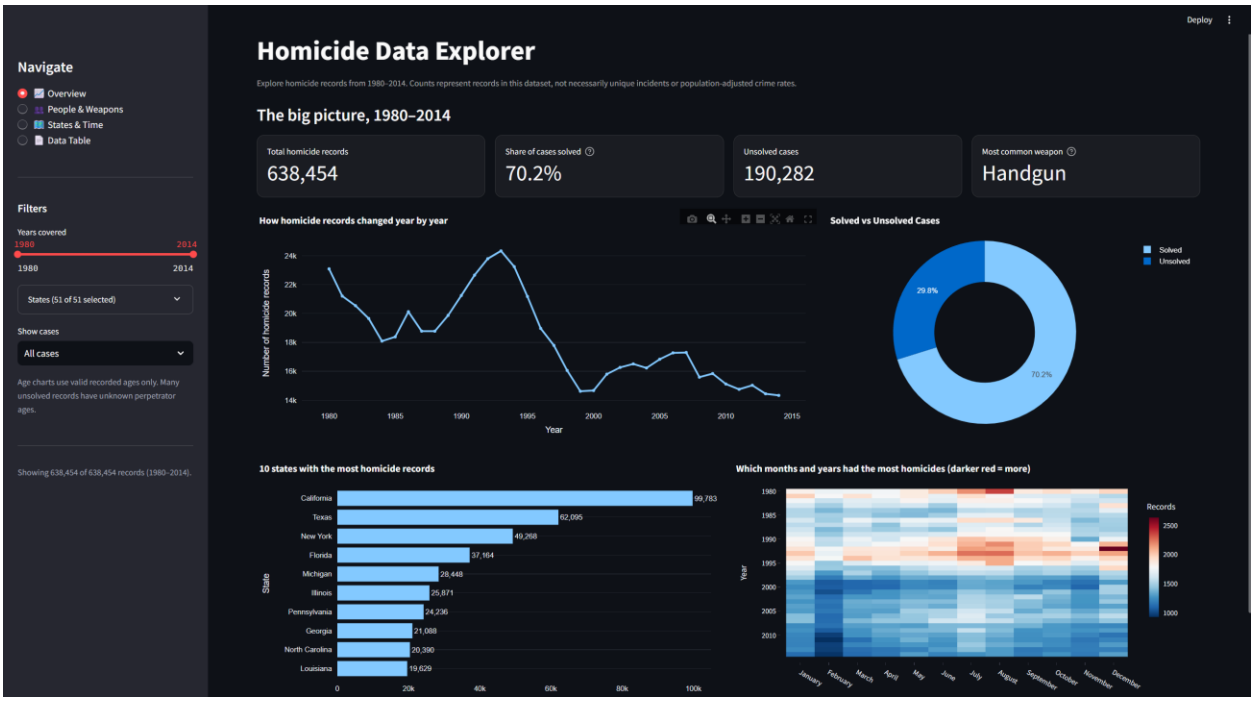


Fig:7

5.2 Dashboard Features

The dashboard was built in Streamlit and uses the same cleaned data as the notebook. A sidebar menu moves between four tabs: Overview, People & Weapons, States & Time, and Data Table. Three filters in the sidebar apply to every chart and table: a year slider (1980–2014), a state dropdown where all 51 states start selected and can be cleared or reselected, and a selector for all, solved only, or unsolved only cases. A caption under the filters shows how many of the 638,454 records match.

The Overview tab opens with four summary cards: total homicide records, the share of cases solved, the number of unsolved cases, and the most common weapon. Below them are the annual line chart, a donut chart of solved versus unsolved cases, the 10 states with the most records, and the month-by-year heatmap. The People & Weapons tab shows perpetrator age in 5-year groups, the age difference between perpetrator and victim, and the eight most common weapons for male and female perpetrators, using only records with valid ages. The States & Time tab ranks states by solved rate (with a switch between the best and worst, limited to states with at least 100 cases), shows a monthly heatmap for any single state, compares solved rates by weapon (at least 500 cases), and lists the 15 states with the most unsolved cases. The Data Table tab shows the first 500 filtered records and lets the reader download all of them as a CSV file.

The charts are interactive: hovering shows exact values, and they can be zoomed, panned, or saved as images. Chart titles state what each chart shows, and some update with the filters, such as the most common perpetrator age group. The dashboard reports recorded counts and solving rates, not population-adjusted rates, so it shows what the records contain rather than why the patterns occur.

6. Insights & Conclusion

6.1 Key Insights

• Annual homicide-record counts peaked in 1993 (24,335) and declined to 14,331 in 2014.
• Of 638,454 records, 448,172 (70.20%) are marked solved and 190,282 (29.80%) are marked unsolved.
• Handgun is the most frequently recorded weapon, with 317,484 records.
• Among state groups with at least 100 records, North Dakota has the highest solving rate (93.18%) and the District of Columbia the lowest (34.31%).
• The 20–24 age group has the highest number of records among perpetrators with valid recorded ages.
• Perpetrator age is coded as 0 in 216,327 raw records; 99.66% of unsolved records have this code, so age-based findings mainly describe cases with a known age and should be interpreted cautiously.

6.2 Conclusion and Recommendations

The analysis shows how homicide-record counts changed over time, how records were distributed across states, and how recorded weapon types and known perpetrator ages relate to the descriptive patterns. The annual count peaked in 1993 at 24,335 and was 14,331 in 2014. Overall, 70.20% of records are marked solved. North Dakota has the highest solving rate (93.18%) and the District of Columbia the lowest (34.31%) among state groups with at least 100 records. Handguns are the most common recorded weapon, while solving rates vary by weapon category. The 20–24 age group has the highest count among records with valid perpetrator ages. These results describe the available records; they do not prove why trends occurred or why one state or weapon category has a different solving rate. A key limitation is that perpetrator age is unavailable for most unsolved records. Other limitations include Unknown categories and the fact that counts are record totals rather than population-adjusted homicide rates. Future work could compare rates using population denominators, investigate missing-data patterns, and validate differences with suitable statistical tests.

7. References

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